Architecture and Design

> **Series:** M2S | **Notebook:** 4 of 8 | **Created:** January 2026 | **Last Updated:** 01/30/2026

Architecture design is critical for a successful migration. The key difference between Managed and SaaS is connectivity—your monitored infrastructure must reach Dynatrace SaaS endpoints.

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Prerequisites

Before starting this notebook, you should have:

Requirement	Description
Completed M2S-01 to M2S-03	Discovery and planning complete
Network diagrams	Current Managed network topology
Firewall access	Ability to request firewall changes
SaaS tenant info	Tenant ID and URL

Learning Objectives

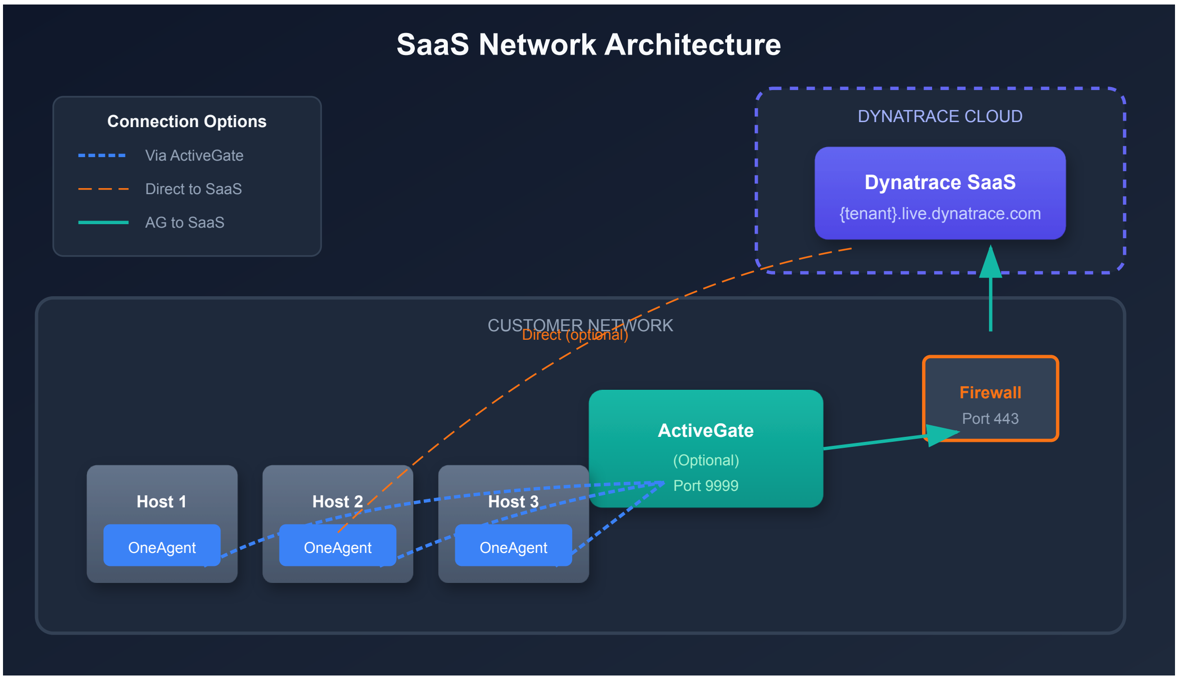
By the end of this notebook, you will:

- Design the network architecture for SaaS connectivity
- Plan ActiveGate deployment for your environment
- Configure security controls for the migration
- Understand high availability considerations

1. Introduction

Key Architectural Changes

Component	Managed	SaaS
Cluster	On-premises servers	Dynatrace-hosted
OneAgent target	Internal cluster	SaaS endpoints or ActiveGate
ActiveGate role	Cluster component	Environment routing/extensions
Network flow	Internal only	Outbound to internet



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2. Network Architecture

2.1 SaaS Endpoints

All communication to Dynatrace SaaS uses HTTPS (port 443).

Endpoint Pattern	Purpose
`{tenant-id}.live.dynatrace.com`	Main SaaS cluster
`{tenant-id}.apps.dynatrace.com`	Apps and platform services

2.2 Connectivity Options

Option	OneAgent →	ActiveGate →	Best For
Direct	SaaS	N/A	Open internet access
Via Proxy	Proxy → SaaS	N/A	Proxy-controlled environments
Via ActiveGate	ActiveGate	SaaS	Restricted networks
Via AG + Proxy	ActiveGate	Proxy → SaaS	Most restricted

2.3 Firewall Requirements

Minimum Requirements:

Source	Destination	Port	Protocol
OneAgent hosts	SaaS endpoint	443	HTTPS
ActiveGate	SaaS endpoint	443	HTTPS

If Using ActiveGate Routing:

Source	Destination	Port	Protocol
OneAgent hosts	ActiveGate	9999	HTTPS
ActiveGate	SaaS endpoint	443	HTTPS

2.4 DNS Resolution

Ensure DNS can resolve:
- `{tenant-id}.live.dynatrace.com`
- `{tenant-id}.apps.dynatrace.com`
- `*.dynatrace.com` (for additional services)

2.5 Network Testing

Test connectivity before migration:

```
# Test HTTPS connectivity to SaaS
curl -v https://{tenant-id}.live.dynatrace.com/api/v1/time

# Test from a monitored host
curl -v https://{tenant-id}.live.dynatrace.com:443
```

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3. Network Zones

Network Zones provide optimized routing of telemetry data from OneAgents to ActiveGates to SaaS. This is especially important when migrating from Managed.

3.1 What Are Network Zones?

Network Zones allow you to:
- Group ActiveGates and OneAgents by network location
- Optimize traffic routing within and between network segments
- Define failover behavior for high availability
- Reduce cross-datacenter traffic

3.2 Why Network Zones Matter for Migration

Consideration	Impact
Existing topology	May need to recreate network zone structure in SaaS
ActiveGate placement	Zones determine which AGs serve which hosts
Failover routing	Hosts fail over to AGs in alternative zones
Bandwidth optimization	Local zones reduce WAN traffic

3.3 Network Zone Planning

Zone Type	Description	Example
Primary	Default zone for a datacenter	`datacenter-east`
Alternative	Failover zone(s)	`datacenter-west`
Cloud-specific	Zone for cloud regions	`aws-us-east-1`
Environment-specific	By environment type	`production`, `nonprod`

3.4 Configuring Network Zones

Step 1: Create Network Zones in SaaS

Navigate to **Settings → Network zones** or use the API:

```
curl -X POST "https://{tenant}.live.dynatrace.com/api/v2/networkZones" \
  -H "Authorization: Api-Token {token}" \
  -H "Content-Type: application/json" \
  -d '{
    "id": "datacenter-east",
    "description": "Primary datacenter in East region",
    "alternativeZones": ["datacenter-west"]
  }'
```

Step 2: Assign ActiveGates to Zones

During ActiveGate installation, specify the network zone:

```
# Linux
sudo /bin/sh Dynatrace-ActiveGate-Linux.sh --set-network-zone=datacenter-east

# Or via configuration after installation
sudo /opt/dynatrace/gateway/config/config.properties
# Add: networkzone=datacenter-east
```

Step 3: Configure OneAgents for Network Zones

```
# During OneAgent installation
sudo /bin/sh Dynatrace-OneAgent-Linux.sh --set-network-zone=datacenter-east

# Or after installation via oneagentctl
sudo /opt/dynatrace/oneagent/agent/tools/oneagentctl --set-network-
zone=datacenter-east
```

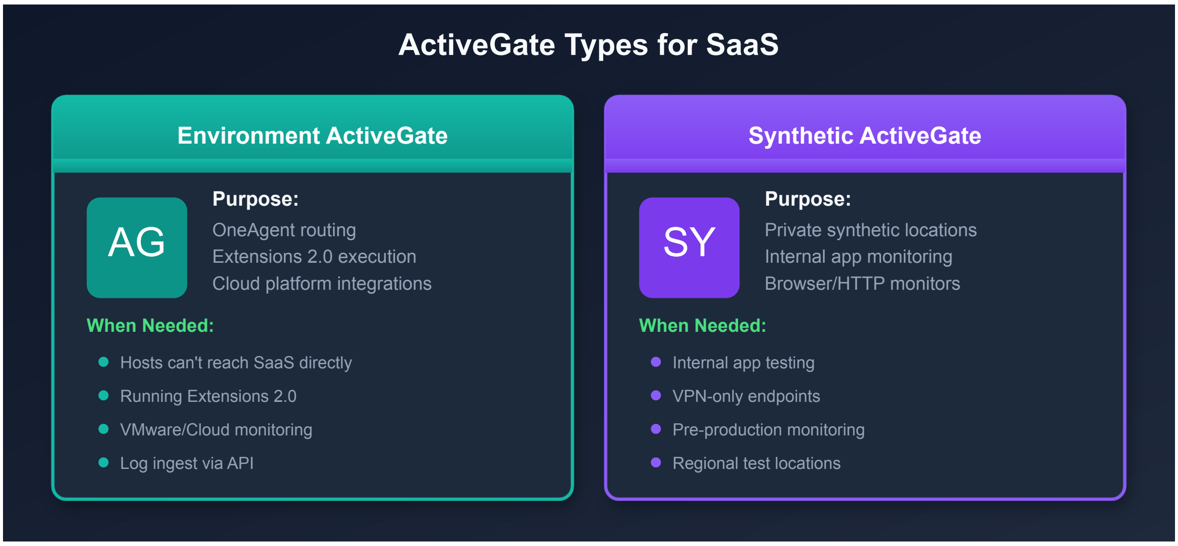
3.5 Network Zone Best Practices

Practice	Rationale
Minimum 2 AGs per zone	High availability within zone
Define alternative zones	Failover if primary zone unavailable
Match physical topology	Zones should reflect network segments
Use meaningful names	Clear identification of location/purpose
Plan before migration	Deploy zones before moving OneAgents

3.6 Migration Consideration: Host Groups vs Network Zones

Feature	Host Groups	Network Zones
Purpose	Logical grouping, config targeting	Traffic routing
Affects	Config inheritance	AG selection
Migration	Export/import via API	Recreate in SaaS
Assignment	Per host	Per host

> **Important:** If you used Network Zones in Managed, you must recreate them in SaaS before migrating OneAgents. Zone configuration does not transfer automatically.



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4. ActiveGate Design

4.1 ActiveGate Types for SaaS

Type	Purpose	When Needed
Environment ActiveGate	OneAgent routing, extensions	Network-restricted hosts
Synthetic ActiveGate	Private synthetic monitoring	Internal application testing

4.2 Do You Need ActiveGate?

Scenario	ActiveGate Required?
Hosts can reach SaaS directly	No (optional for extensions)
Hosts behind firewall, no direct internet	Yes, for routing
Running Extensions 2.0	Yes
Private synthetic monitoring	Yes
VMware monitoring	Yes

4.3 ActiveGate Sizing

Hosts Routed	CPU Cores	Memory	Disk
Up to 500	2	4 GB	20 GB
500-1,500	4	8 GB	40 GB
1,500-5,000	8	16 GB	80 GB
5,000+	Multiple AGs	Scale horizontally	-

4.4 ActiveGate Placement

Best Practices:

1. **Close to monitored hosts** - Minimize latency
2. **In each network zone** - One AG per isolated segment (see Section 3)

3. **High availability** - Deploy minimum of 2 per datacenter
4. **DMZ for external access** - If needed for cloud integrations

4.5 ActiveGate Groups

Use groups to organize ActiveGates:

Group	Purpose	Members
`production-routing`	Prod OneAgent routing	AG-PROD-01, AG-PROD-02
`nonprod-routing`	Non-prod routing	AG-NONPROD-01
`extensions`	Extension execution	AG-EXT-01
`synthetic`	Private synthetic	AG-SYNTH-01, AG-SYNTH-02

4.6 Install New ActiveGates in Parallel

> **Best Practice:** Install new SaaS-connected ActiveGates in parallel with existing Managed ActiveGates before migrating OneAgents.

This approach:
- Validates connectivity before impacting monitored hosts
- Allows gradual migration of OneAgents
- Provides easy rollback if issues occur
- Maintains monitoring continuity during transition

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5. Security Architecture

5.1 API Token Strategy

Create purpose-specific tokens:

Token Purpose	Required Scopes
OneAgent installation	`InstallerDownload`
ActiveGate installation	`InstallerDownload`
Configuration migration	`settings.read`, `settings.write`
Entity queries	`entities.read`
Metrics queries	`metrics.read`
Log queries	`logs.read`
Automation	`automation.read`, `automation.write`

5.2 User Access Migration

Authentication	Managed	SaaS
Local users	Managed cluster	Account Management
SAML/SSO	IdP → Managed	IdP → Dynatrace Account
LDAP	Direct LDAP	Not supported (use SAML)

5.3 Permission Mapping

Managed Role	SaaS Equivalent
Cluster admin	Account admin
Environment admin	Environment admin
Monitor user	Viewer + specific policies
Custom roles	IAM policies

5.4 Network Security

Control	Implementation
TLS encryption	Enforced by SaaS (TLS 1.2+)
Certificate pinning	OneAgent validates SaaS certs
IP allowlisting	SaaS IPs published by Dynatrace
Egress filtering	Limit to Dynatrace domains only

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6. High Availability

6.1 SaaS HA (Provided by Dynatrace)

Dynatrace SaaS includes:
- Multi-region deployment
- Automatic failover
- Data replication
- 99.5% SLA (varies by contract)

6.2 Customer-Side HA

ActiveGate High Availability:

Component	HA Strategy
Environment AG	Deploy multiple per network zone
Synthetic AG	Deploy pairs at each location
Load balancing	OneAgent auto-discovers available AGs

6.3 Failover Behavior

Scenario	OneAgent Behavior
Primary AG unavailable	Fails over to secondary AG
All AGs unavailable	Buffers data locally (up to 2 hours)
AG + SaaS available	Resumes normal transmission

Architecture Design Checklist

Design Area	Completed
Network connectivity verified	[ ]
Firewall rules documented	[ ]
Network Zones planned	[ ]
ActiveGate placement planned	[ ]
ActiveGate sizing calculated	[ ]
API token strategy defined	[ ]
User migration planned	[ ]
HA requirements addressed	[ ]
Bandwidth impact assessed	[ ]

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7. Next Steps

Immediate Actions

1. **Finalize network design** - Document all connectivity paths
2. **Request firewall changes** - Submit change requests early

3. **Plan Network Zones** - Recreate any zones from Managed
4. **Size ActiveGates** - Based on hosts to route
5. **Plan token strategy** - Document required scopes
6. **Create architecture diagram** - Visual documentation

Continue the Series

Next Notebook	Focus
M2S-05: Configuration Migration	Settings, dashboards, and automation

Architecture Resources

- ActiveGate Documentation
- Network Zones
- Network Requirements
- OneAgent Communication

Summary

In this notebook, you learned:

- How to design network architecture for SaaS connectivity
- Network Zones for optimized traffic routing
- ActiveGate sizing and placement strategies
- Security considerations for the migration
- High availability patterns for customer-side components

> **Key Takeaway:** The primary architectural change is network connectivity. Ensure all monitored hosts can reach SaaS endpoints, either directly or through ActiveGates. Plan Network Zones before migrating to optimize traffic routing.

Continue to M2S-05: Configuration Migration for guidance on migrating settings and configurations.
